

FACULTY OF ENGINEERING

INGB417 – GROUP ASSIGNMENT

Group 9

Submitted to: Mrs Willemien Kruger

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Student Name	Student Number	Contributions
Sheldon Farmer	37109944	33.33%
Ofentse Mathebula	34897658	33.33%
Karla van Dyk	37511459	33.33%

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1. Executive Summary

A team of industrial engineers has been appointed as facility designers to design a new facility for Wiring.com. The facility has to be designed from scratch and make use of sustainable energy solutions. The factory is required to make five different types of netting with four different processes. A 10% expansion must also be considered within the next few years.

2. Background

Wiring.com is a company that specializes in the manufacturing of wire-based products specifically netting products like bird netting, chicken netting, jackal netting, pig netting, welded mesh. They have been experiencing an increase in demand for their products however their current facility does not have adequate capacity to meet the demand from customers which results in production constraints at the current facility.

Due to not having adequate capacity in their current facility and not being able to purchase the properties adjacent to their current facility, they made a strategic decision to purchase a new piece of land that spans over an area of 50 HA, to build an entirely new facility from the ground up as there is currently no buildings or previously developed facilities on the property that they purchased. The property is located in Sedibeng District in Gauteng about 10km away from their current facility.

The purpose of the project to develop a new facility is to design a facility with an optimal layout and that increases efficiency and most importantly, is able to accommodate the increased production capacity. Along with this, the facility must have sufficient space for all manufacturing equipment and -machinery to ensure that the facility can accommodate the number of machines that will be required to meet demands per day as well as the total annual demand.

3. Design Problem and Objectives

3.1. Design Problem Statement

Wiring.com's current facility cannot meet the rising demand for its netting products, requiring a new manufacturing plant on a 50-hectare site in Sedibeng, Gauteng. The facility must accommodate the full production process, optimize material handling and storage, support personnel needs, incorporate sustainable energy solutions, and allow for future expansion while ensuring efficiency and compliance with operational constraints.

3.2. Design Objectives

- Develop an optimized facility layout that ensures efficient material flow, space utilization, and safety.
- Select and recommend suitable equipment for wire drawing, galvanization, netting production, and packaging to meet production demand efficiently.
- Design an effective material handling system with proper storage allocation for raw materials, work-in-progress, and finished goods.
- Plan for personnel and support facilities including offices, restrooms, locker rooms, cafeteria, and parking while ensuring compliance with industry standards.
- Integrate sustainable energy solutions and allow for a 10% expansion capacity to accommodate future growth.

- Optimize logistics and distribution by designing efficient receiving and dispatch areas, internal roadways, and parking spaces.

4. Design Requirements

- 50HA plot of land
 - Max 37.5HA of land may be for facility
 - 12.5HA allocated to roads, parking and green buffer zones
- No requirements on shape of land plot or facility
- Space minimized to reduce costs
- Sufficient parking for all personnel
- Lockers for all personnel
- Rest rooms in accordance with standards
- Offices for plant management, operations supervisor, sales, purchasing, finance and inventory management departments, administration personnel
- Reception area for small order pick-ups, paperwork and deliveries
- Storage rooms for engineering, operations, maintenance and spare parts
- 7 days storage for small orders and 2 days storage for large orders
- Storage for 4% scrap produced during drawing operation
- Raw material storage for 3 days' demand
- Consider 10% expansion in the next 5-10 years
- Small orders are defined as an order of 1 ton or less and makes up 10% of total production
- A working day shift is 8 hours long, but all machines only run for 7 hours to account for setup, cleaning and small maintenance that may need to be done.

5. Design Assumptions and Constraints

5.1.Process Flow

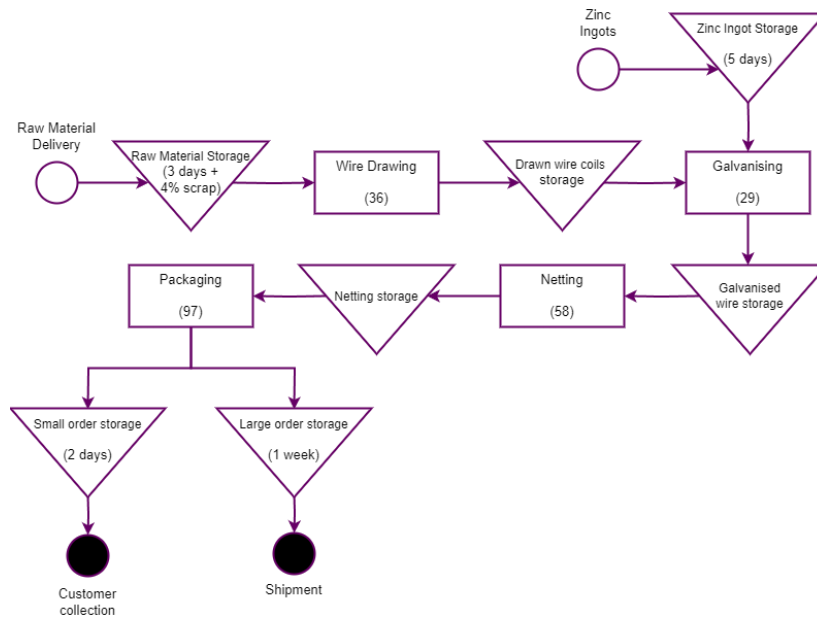


Figure 5.1.1: High-level Process Flow Diagram

5.2.Current State and Given Information

Operation A (Wire Drawing)

- Steel wire coils used
 - 5mm diameter
 - 7000m length
- Drawn wire rolled into coils before galvanization

Table 5.1.1: Wire diameters of drawn wire

Product	Wire diameter (mm)
Bird Netting	0.6
Chicken Netting	0.9
Jackal Netting	1.8
Pig Netting	1.8
Welded Mesh	2.5

Operation B (Galvanising of wire)

- Wire is drawn through bath with liquid
- Zinc ingots bought to keep process going

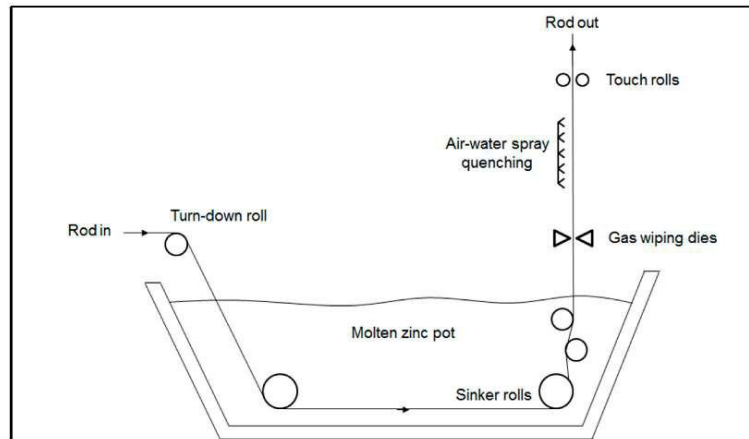


Figure 5.1.1: Concept Diagram of Galvanisation Process

Operation C (Netting Production)

- Total annual demand of 26 000 tons

Table 5.1.2: Product Specifications

Product	Roll Specifications				Aperture (mm)	Annual Demand (tons)
	Height (m)	Length (m)	Diameter (m)	Weight (kg)		
Bird Netting	0.9	50	0.25	22.5	13	4000
Chicken Netting	1.8	50	0.28	41.9	25	5500
Jackal Netting	0.6	50	0.37	19.5	75	2500
Pig Netting	0.9	30	0.29	14.7	90	8000
Welded Mesh	2	30	0.34	70	50	6000

Operation D (Packaging)

No specifications were given for packaging. Research done about this and decisions made about which machines to use can be seen in section 6.5 ([Operation D \(Packaging\)](#)) of this document.

6. Detail Design Specification

6.1.Raw Material Storage

The raw material of this process is steel wire coils of 7km long, weighing 2.5 tons and with a diameter of 5mm. It is assumed that the diameter of a coil is 1m.

The total volume of wire in the coils is calculated from the demand. that A total of three days storage space is needed for production. 148 coils are needed to meet this demand. Coils are stored in racks of six coils per rack as can be seen in figure 6.1.1 below. These racks will have wheels with locks on them for easy movement of the racks when attached to vehicles and locked when needed to stay in place. The storage racks have dimensions of 4m*1.2m*1.2m (l*w*h).



Figure 6.1.1: Raw Material Storage Racks

The total area needed for one rack (including walkway around it) is about 9m². A total of 25 racks will be needed to satisfy the storage requirements as 6 coils can be stored on each rack. The 25 racks are stored in a grid of 5 by 5 racks. The total area can be calculated by multiplying the amount of racks and the area of each rack, required area for raw materials storage for three days is 240m².

6.2.Operation A (Drawing)

Specifications



Figure 6.2.1: Model 450 Drawing Machine

Figure 6.2.1 above shows drawing machine model 450, used to draw the wires from 5 mm to 1.8 mm for jackal netting and pig netting and to 2.5 mm for welded mesh. This same machine is also used to draw wires from 5 mm to 1.8 mm for bird netting and chicken netting before entering drawing machine model 300 to be drawn even thinner as drawing machine model 450 has a minimum diameter of 1 mm it can draw a wire. Figures 6.2.2 and 6.2.3 below show the drawing machine Model 300, used to draw the wires to 0.6 mm and 0.9 mm for bird netting and chicken netting.



Figure 6.2.2: Model 300 Drawing Machine Input



Figure 6.2.3: Model 300 Drawing Machine Output

Table 6.2.1: Drawing Machine Specifications

Model 300	Model 450
Max Diameter in: 2.8 mm	Max Diameter in: 5mm
Min Diameter out: 0.5 mm	Min Diameter out: 1 mm
Max speed: 25 m/s	Max speed: 16 m/s
Dimensions: 2,4m * 1,1m * 1,1m	Dimensions: 6m * 1,5m * 2m
Number of operators: 1	Number of operators: 1
Price: \$ 42870,00	Price: \$ 50000,00

These machines are highly automated and only need one operator per machine to oversee that the machine works smoothly as well as to input new raw material. These machines also save a lot of labour costs.

Calculations

To determine the total kilometres that have to be produced by the drawing process, the length of a coil after being drawn to the different diameters was calculated by equating the volume of the coil before drawing and after drawing. In the equation below r_i and L_i are the radius of the raw material coils and length of raw material coil, respectively. Whereas r_f and L_f are the radius and length of a coil after the drawing process.

$$\pi r_i^2 L_i = \pi r_f^2 L_f$$

The weight per meter of the different thickness of wire needed is then calculated by dividing the weight of a coil (2.5 ton) by the length of the coil after being drawn. The total metres needed of drawn wire to produce one roll is calculated by weight per roll by the weight per meter. The

amount of rolls needed per annum is then calculated by dividing the demand per annum by the weight per roll. The amount of rolls needed per annum is then multiplied by the meters per roll needed and then divided by 220 (the amount of working days per annum) to get the kilometres of drawn wire needed per day. A scrap allowance of 4% is then also added to compensate for expected scrapping of materials during this process. The results (rounded) from these equations can be seen in table 6.2.1 below.

Table 6.2.1: Drawing Process Calculations Results

Process	Bird Netting	Chicken Netting	Jackal Netting	Pig Netting	Welded Mesh	Totals
Wire radius (m)	0.0003	0.00045	0.0009	0.0009	0.00125	-
Length of coil after drawing (m)	486111	216049	54012	54012	28000	838185
Weight per meter (kg)	0.0051	0.0116	0.0463	0.0463	0.0893	-
Metres needed per roll	4375	3621	421	318	784	-
Rolls needed per annum	177778	131265	128205	544217	85714	-
Drawn wire needed per annum (km)	777778	475309	54012	172840	67200	1547138
Drawn wire needed per day with 4% scrap allowance (km)	3683	2251	256	818	318	7325

It is decided to not run the machines at full speed in order to decrease the wear and tear on the machines and the wires. The drawing speeds used can be seen in table 6.2.3 below. Detailed calculations and unrounded values of all calculations can be seen in Appendix A.

From here the number of drawing machines is calculated by taking the total kilometres of wire needed after drawing and dividing it by the total kilometres a drawing machine can draw in one working day. It is important to note that for the bird and chicken netting the wire goes through two drawing machines. One machine is used to take the wire diameter from 5mm to 1mm and another is used to further reduce the diameter to 0.6mm and 0.9mm as needed. From these calculations, it is evident that 36 drawing machines is needed to keep up with demand. A summary of the calculation results can be seen below in table 6.2.3.

Table 6.2.3: Drawing Machine Calculations

Process	Bird Netting	Chicken Netting	Jackal Netting	Pig Netting	Welded Mesh	Totals
Drawn wire needed per day with 4% scrap allowance (km)	3683	2251	256	818	318	7325
Machine 1 drawing speed (m/s)	15	15	20	20	20	-
Machine 2 drawing speed (m/s)	15	15	-	-	-	-

Length one machine draws (m)	378000	378000	504000	504000	504000	2268000
No. of machines needed	20	12	1	2	1	36

Workstation Layout

The workstation layout for the different drawing machines can be seen below in figures 6.2.4 and 6.2.5. Figure 6.2.4 indicates the layout for the jackal netting, pig netting and welded mesh. Figure 6.2.5 show the layout for the bird and chicken netting where two machines are needed to draw the desired diameter. A 1m walkway is indicated in yellow all around the machine for use by operators. All machines pull the wire to be drawn directly from the raw material storage.

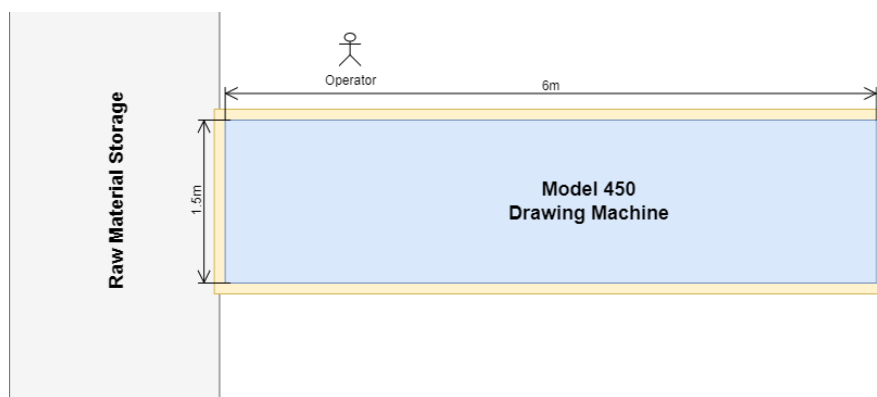


Figure 6.2.4: Welded Mesh and Jackal & Pig Netting layout

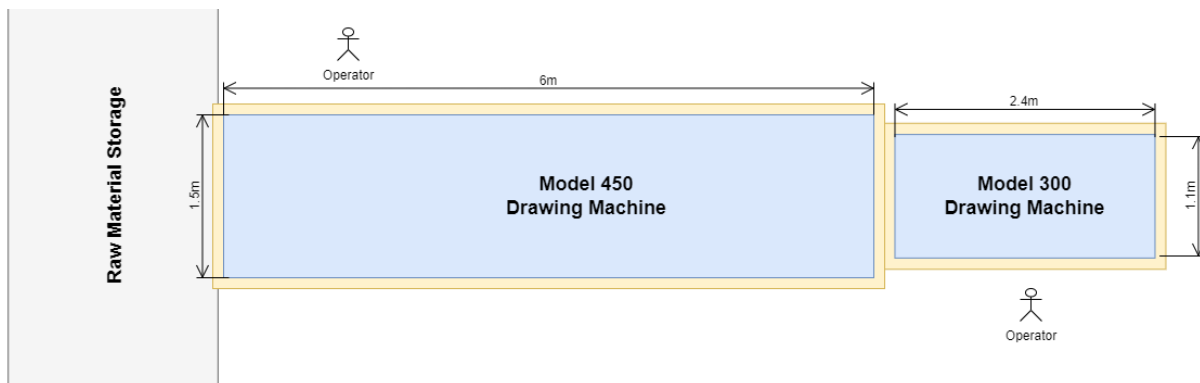


Figure 6.2.5: Bird & Chicken Netting layout

6.3.Operation B (Galvanising)

Galvanising

Specifications



Figure 6.3.1: Wire coiling and de-coiling



Figure 6.3.2: Zinc Ingot input section



Figure 6.3.3: Hot-dip galvanisation



Figure 6.3.4: Wire cooling

Table 6.3.1: Galvaniser Specifications

Dimensions (l*w*h)	100*5*3 (m)
Operators	Four
Throughput	600 kg/h
Voltage	380V 3P 50Hz
Price	R700 000
Zinc Coating Thickness	50 microns

Calculations

To determine the number of galvanisers needed, the hourly weight requirement of galvanised wire is calculated and divided by the throughput rate of 600kg/h. A rounded summary of these

calculations is seen in table 6.3.2 below. Detailed calculations and unrounded values of all calculations can be seen in Appendix A.

Table 6.3.2: Galvaniser Calculations

Product	Annual demand (kg)	Daily Zinc needed (kg)	Machines needed
Bird Netting	4 000 000	19 169	2
Chicken Netting	5 500 000	17 121	2
Jackal Netting	2 500 000	3 789	1
Pig Netting	8 000 000	12 124	1
Welded Mesh	6 000 000	6 497	1
Total	26 000 000	58 700	7

Zinc Ingot Storage

Specifications

Zinc ingot slabs of 25kg each are used for the galvanisers (Boliden, 2022). The slabs are delivered and stored in pallets that consist of 10 layers with 8 zinc ingot slabs on each layer as can be seen in figure 6.3.5 below. One work week's (5 days) storage is needed for the zinc ingots.



Figure 6.3.5: Zinc Ingot Pallet

Calculations

To calculate the total weight of Zinc-ingots used daily, the total surface area of all wire that must be coated is divided by the galvanised coating weight per square meter. The rounded calculation results can be seen in table 6.3.3 below.

Detailed calculations and unrounded values of all calculations can be seen in Appendix A.

Table 6.3.3: Zinc Ingot Calculations

Product	Daily demand (m)	Wire Radius (m)	Surface Area (m ²)	Galvanised Coating (kg/m ²)	Zinc needed daily (kg)
Bird Netting	3682660	0.0003	6942	0.3	2082
Chicken Netting	2250514	0.00045	6363	0.3	1909
Jackal Netting	255740	0.0009	1446	0.3	435
Pig Netting	818369	0.0009	4628	0.3	1388
Welded Mesh	318182	0.00125	2499	0.7	1749
Total	7325465		21878		7563

From here it is calculated that 303 zinc ingots are needed daily and 1513 are needed for one work week. Using the dimensions of one pallet (1m*0.5m*0.4m) it is calculated that 10m² of storage area is needed per galvaniser for one week.

Workstation Layout

The workstation layout for the galvanisers and zinc storage can be seen below in figure 6.3.6. A 1m walkway is indicated in yellow all around the machine for use by operators. It is also important to note that the zinc ingot storage is large enough so that it will also be used by the galvaniser put on the other side of the storage. There will thus be two galvanisers for each zinc ingot storage area.

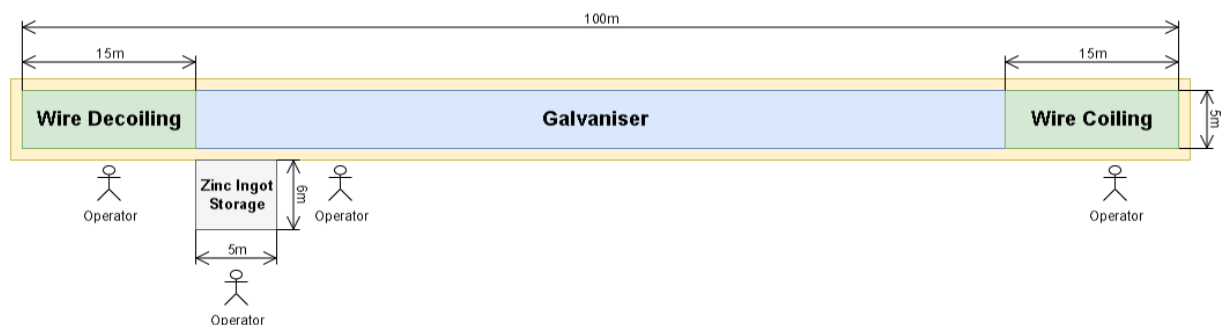


Figure 6.3.6: Galvaniser & Zinc Ingot storage layout

6.4.Operation C (Netting)

Specifications for Hexagonal Wire Netting Machine



Figure 6.4.1: Netting Machine Input



Figure 6.4.2: Netting Machine Output

Figure 6.4.1 above shows where the drawn and galvanized steel coils will be placed as input to the hexagonal wire netting machine. Figure 6.4.2 shows the output that the hexagonal wire netting machine produces.

Table 6.4.1: Hexagonal Wire Netting Machine Specifications

Description	Value
Max Netting (mm) Width:	4500
Max Diameter of Weaving Wire (mm)	2.7
Production Capacity (Sq. Meter / hour)	500
Dimensions (L*W*H)	15*6*3
Number of Operators	4

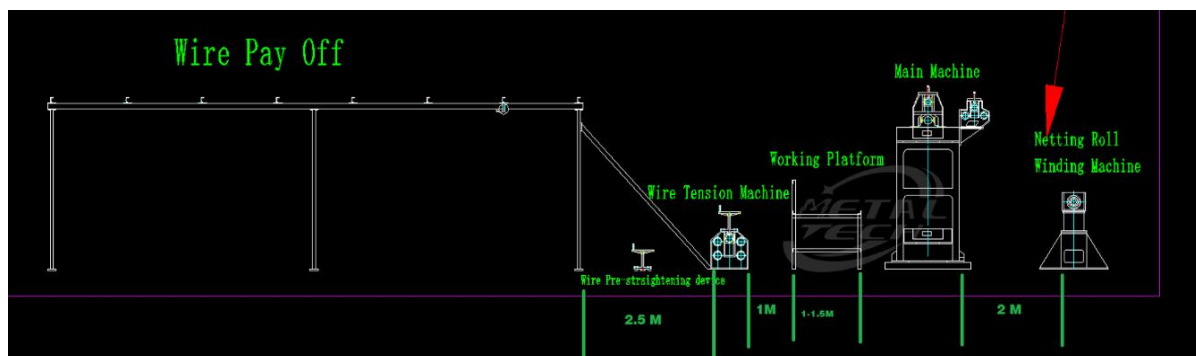


Figure 6.4.3: Netting process flow

The hexagonal wire netting machine's settings can be adjusted to the required aperture sizes by making adjustments to the twisting and spacing mechanisms of the machine.

Calculations for Hexagonal Wire Netting Machine

To determine how many hexagonal wire netting machines will be required to meet demand, we first had to determine how many rolls of each product were going to have to be produced per day.

Table 6.4.2: Netting Calculation Results

	Roll Length (m)	Roll Hight (m)	Roll Weight (kg)	Customer demand (Tons per Year)	Customer Demand (kg per Year)	Rolls per Year	Rolls per Day (Demand)
Bird Netting	50	0,9	22,5	4000	4000000	177778	808
Chicken Netting	50	1,8	41,9	5500	5500000	131265	597
Jackal Netting	50	0,6	19,5	2500	2500000	128205	583
Pig Netting	30	0,9	14,7	8000	8000000	544218	2474
Welded Mesh	30	2	70	6000	6000000	85714	390

From table 6.4.2 above we see that the weight of each product roll was taken and divided by the total annual demand in weight for each product to obtain the number of rolls that had to be produced per year. Then the number of annual rolls was further divided by the number of operational days in a year (220 days) and we then obtained the number of rolls that had to be produced per day for each product.

After doing research on the hexagonal netting machine's specifications, we found that this machine can net at a maximum rate of 500 square meter per hour (as indicated above). We don't however want to use this machine at its full capacity; hence we made the decision to have it operate at 450 square meter per hour.

Table 6.4.3 Netting Machine Calculations

	Rolls per Day	Area per Roll (m ²)	Total Net Area per Day Needed (m ²)	Machine Production Rate (m ² / h)	Machine Production Rate (m ² / day)	Number of Netting Machines Needed
Bird Netting	808	45	36364	450	3600,00	10
Chicken Netting	597	90	53699	450	3600,00	15
Jackal Netting	583	30	17483	450	3600,00	5
Pig Netting	2474	27	66790	450	3600,00	19
Total						49

From table 6.4.3 above we see that the area of a roll of each respective product was obtained by multiplying the roll height (that was take as the width) with the length of the roll. After obtaining the area of each roll, we calculated the required area to be produced per day to satisfy demand. We thus, multiplid rolls per day required with the area per roll for each respective product. Now that we have the total area to be netted per day for each product, we can now take the machines production rate of 450 square meter per hour to determine the production rate in square meters per day. The total netting area needed per day is then divided by the production rate per day to determine the number of netting machines that is needed in the new facility.

Specifications for Automatic Wire Mesh Welding Machine



Figure 6.4.4: Galvanised coils fed into Netting Machines



Figure 6.4.5: Automatic Wire Mesh Welding Machine Output

Table 6.4.4: Automatic Wire Mesh Welding Machine Specifications

Description	Value
Production Rate (times/min)	50
Machine Dimensions (L*W*H)	4,5m * 2,3m * 1.8m
Max Width of Mesh (m)	2,1m
Number of Operators	3

For the Automatic Wire Mesh Welding Machine, the specification for the production rate is specified slightly different. This machine's production rate was given in terms of times per minute, referring to the number of rows of wire that the machine welds per minute.

Calculations for Automatic Wire Welded Mesh Machine

Table 6.4.4: Automatic Wire Mesh Welding Machine Production Capacity

	Welded Mesh Production Rate (Times/min)	Aperture (m)	Production Rate (m/min)	Production Rate (m/h)	Production Rate (m ² / h)	Welded Mesh Production Rate (m ² / day)
Welded Mesh	50	0,05	2,5	150	300	2400

From table 6.4.4 we see that the number of times per minute was multiplied with the aperture to obtain the length of the roll that the machine can produce in meters per minute. It was then converted to meters per hour. This length was multiplied with the height of the roll (2m) to obtain the total area in square meters per hour that the automatic wire mesh welding machine can produce. This production rate was then converted to square meters per day assuming an 8-hour workday.

Table 6.4.5: Number of Automatic Wire Mesh Welding Machines needed

	Rolls per Day	Area per Roll (m ²)	Total Net Area per Day Needed (m ²)	Machine Production Rate (m ² / h)	Machine Production Rate (m ² / day)	Number of Wire Mesh Welding Machines Needed
Welded Mesh	390	60	23377	300	2400	10

From table 6.4.5 above we see that total number of rolls required per day for welded mesh was multiplied with the area per roll to obtain the total net area per day needed. This total was then divided by the machine production rate in square meters per day to obtain the number of automatic wire mesh welding machines that will be required to meet demand.

Workstation Layouts for Hexagonal Wire Netting Machine

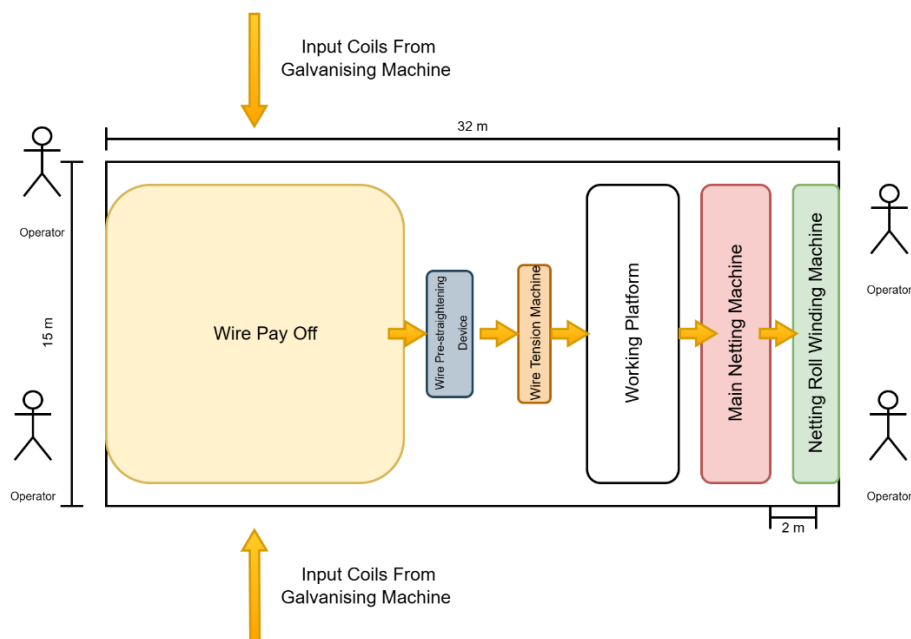


Figure 6.4.6: Hexagonal Wire Netting Machine Workstation Layout

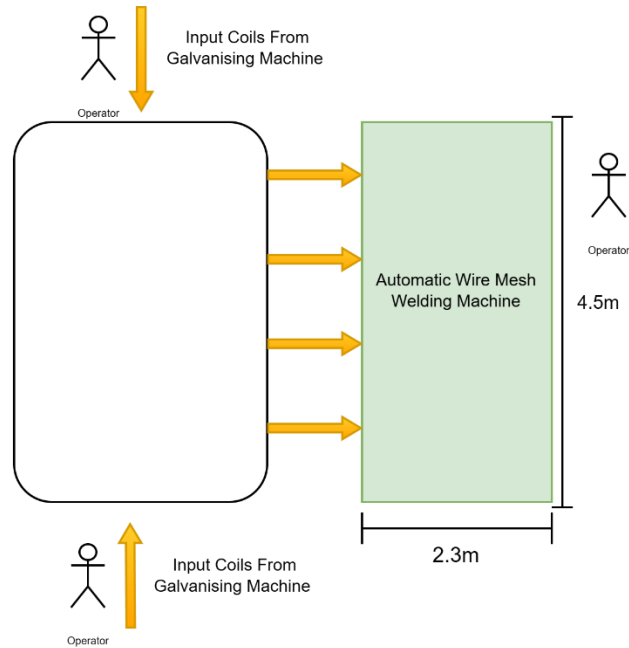


Figure 6.4.7: Automatic Wire Mesh Welding Machine Workstation Layout

6.5.Operation D (Packaging)

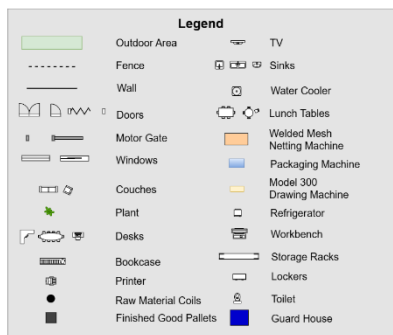
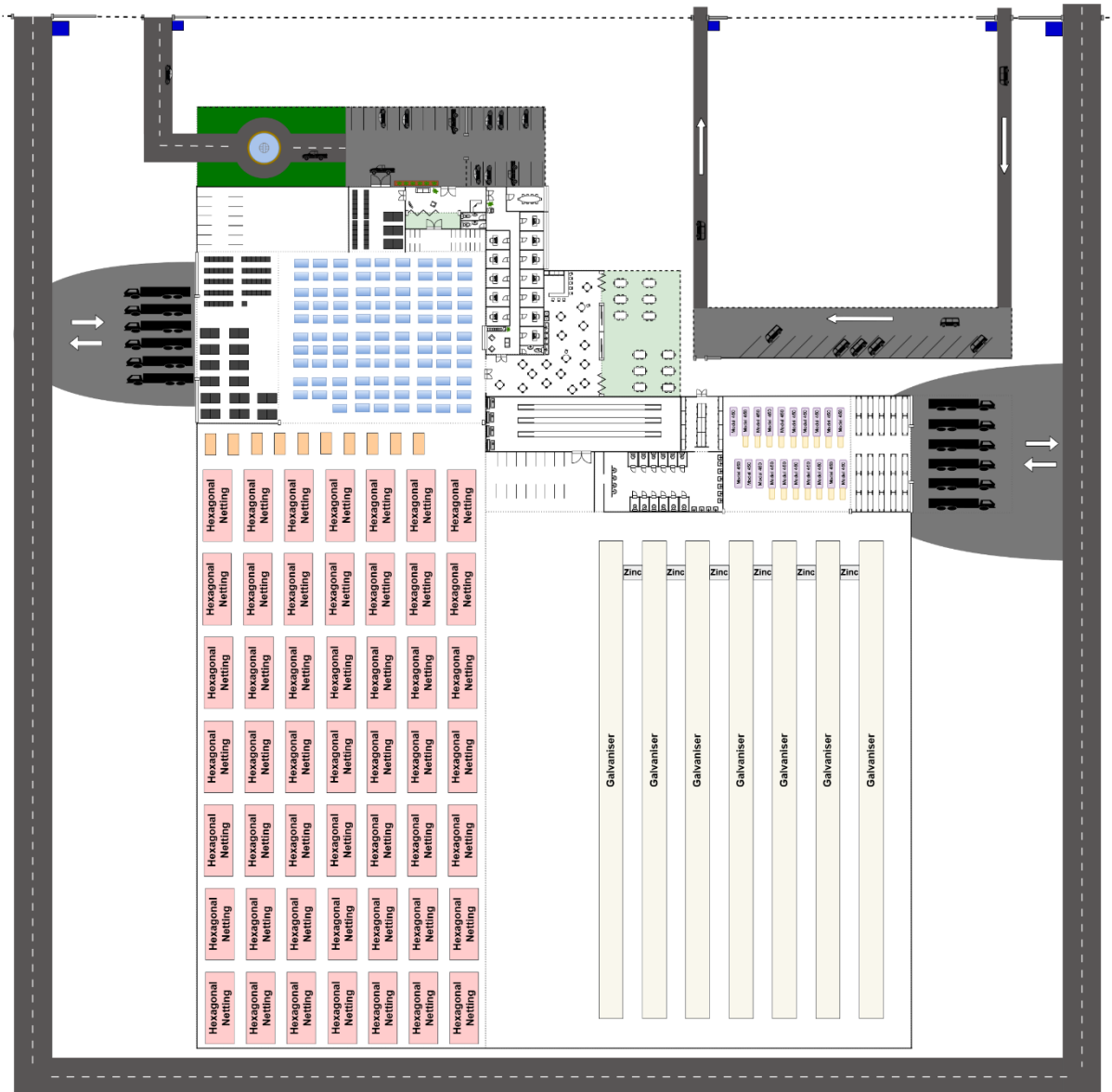
Specifications

The packaging of the finished goods will be the same for all the products; Bird netting, chicken netting, Jackal netting, pig netting and welded mesh. The finished products are packaged using a wire mesh roll packing machine, that uses a High-Density Polyethylene (HDPE). The plastic protects the products from getting damaged during storage and transportation. Every machine requires 1 operator. After wrapping the wire mesh, it is stored on Pallets. A single pallet carries a maximum of 1000 kg. This is made to accommodate customers with small orders that can carry their products on the back of their vans. The image below depicts the packing machines used for packaging.



Figure 6.5.1: Packaging Machine

Master Facility Plan for Wiring.com

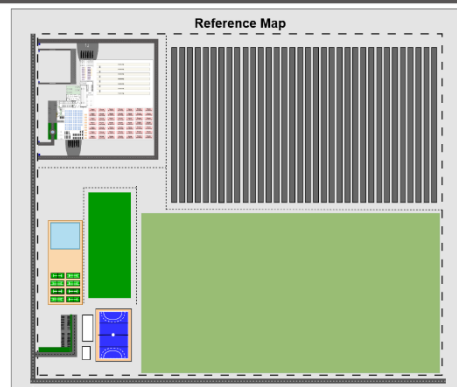


Scale

10m

Total Area 22 500m²
(excl. roads, delivery & loading bays)

Group 9:
Sheldon Farmer
Ofentse Mathebula
Karla van Dyk



Calculations

The packaging process is a critical step in ensuring that the finished netting products are properly wrapped and prepared for distribution. The table below outlines key operational metrics, including the number of rolls processed per day, packaging efficiency, required packaging machines, and the space and personnel needed for each product type. With a

packaging efficiency of 50 rolls per hour, the facility operates multiple machines to meet daily production demands. The total number of packaging machines required is 97, occupying a combined space of 434.63 square meters. Additionally, 97 personnel are allocated to operate the machines and manage the packaging workflow. This data provides a basis for optimizing packaging operations to improve efficiency and reduce space utilization.

Table 6.5.1: Space Requirements for Packaging

	Rolls per Day	Packaging Efficiency (Rolls/hour)	Number of Packaging Machine	Machine Dimension (square meters)	Number of Personnel	Total Space (square meters)
Bird Netting	808,08	50	16	4,48	16	72,40
Chicken Netting	596,66	50	12	4,48	12	53,46
Jackal Netting	582,75	50	12	4,48	12	52,21
Pig Netting	2473,72	50	49	4,48	49	221,65
Welded Mesh	389,61	50	8	4,48	8	34,91
Totals			97		97	434,63

Workstation Layout

The workstations of the packaging area are organized in a group cell format. Where all the machines working on the chicken netting are grouped together and similarly with the other machines for other types of products. This is done to increase the efficiency of each product. The image below depicts the layout of the packaging area.

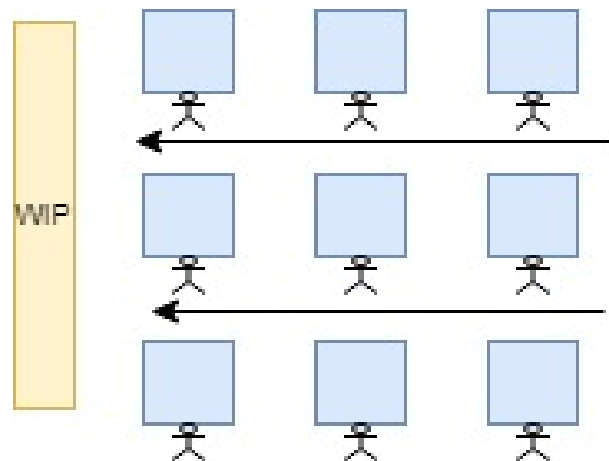


Figure 6.5.3: Packaging Workstation Layout

6.6.Finished goods storage & distribution

Small Orders

Small orders are defined as orders of 1 ton or less and make up 10% of total production. A week's storage space is needed for small orders. Assuming that rolls are stacked on each other in pallets of 16 (four layers of four rolls each) and four pallets are stacked on top of each

other, the weekly storage space can be calculated by multiplying the rolls per day with the area needed and dividing that by four. calculations can be seen in the table below. A week is also assumed to be 7 days. It is also then assumed that each pallet, including walking space around it has an area of 2.5m². Calculations can be seen in Appendix A.

Table 6.6.1: Small Order Storage Area Calculations

Product	Rolls produced per day	Area needed for one roll (m ²)	Total daily area needed for stacks of rolls (m ²)	Weekly area needed for storage (m ²)
Bird Netting	81	0.225	5	32
Chicken Netting	60	0.504	8	53
Jackal Netting	59	0.222	3	23
Pig Netting	248	0.261	16	113
Welded Mesh	39	0.68	7	46
Totals	487	-	39	267

Large Orders

The same calculations and assumptions as above can now be used for the large orders storage area. There however only needs to be enough space for two days storage.

Table 6.6.2: Large Order Storage Area Calculations

Product	Rolls produced per day	Area needed for one roll (m ²)	Total daily area needed for stacks of rolls (m ²)	Total area needed for storage (m ²)
Bird Netting	728	0.225	41	82
Chicken Netting	537	0.504	68	135
Jackal Netting	525	0.222	29	58
Pig Netting	2227	0.261	145	291
Welded Mesh	351	0.68	60	119
Totals	4366	-	343	685

Delivery Trucks

The trucks that are we are going to use to deliver the finished product is a Volvo FH-16, which can carry a capacity of 70 tons in one load. The image below depicts the truck that is used.



Figure 6.6.1: Delivery Truck

6.7. Management Policies

Manufacturing wire mesh products such as chicken netting, pig netting, jackal netting, and welded wire mesh requires strict adherence to safety protocols and factory standards to ensure both worker safety and product quality. The factory layout must accommodate proper safety measures, including workroom doors that are at least 1.2 meters wide and 2 meters high. Floors should be covered with rubber sheets no less than 3 millimeters thick to prevent slips and electrical hazards. Maintaining cleanliness in work areas is essential, with strict housekeeping practices to keep pathways clear and address material spillages promptly.

To prevent workplace accidents, machine guarding is critical. Protective guards must be installed on all moving machinery to shield operators from potential injuries. Electrical safety is another priority, requiring the use of flameproof conduits and the avoidance of non-conductive handle lamps to minimize electrical hazards. Additionally, workers must be equipped with appropriate personal protective equipment (PPE), including gloves, aprons, and breathing apparatuses, particularly in areas where exposure to hazardous materials or fumes is a concern.

By implementing these safety protocols and adhering to recognized industry standards, factories can ensure efficient and secure production environments while delivering high-quality wire mesh products.

6.8. Sustainable Energy Solutions

Solar Power with Battery Storage

Solar energy is one of the most practical and sustainable solutions. By installing solar panels on the facility's roof or surrounding land, the facility can generate a significant portion of its electricity needs from renewable sources. Steel netting production involves energy-intensive processes such as wire drawing, hot-dip galvanising and cutting, making energy costs a major operational factor. Solar power can offset these costs by providing a steady supply of electricity during daylight hours. Additionally, integrating battery storage ensures that excess solar energy generated during the day can be stored and used during peak demand periods or when it is loadshedding, thus increasing energy reliability. Beyond cost savings, solar power reduces dependence on the conventional grid, making the facility more resilient to fluctuations in electricity prices.

On-Site Wind Energy (Small-Scale or Hybrid System)

Wind energy presents another viable option, particularly if the facility is located in a region with sufficient wind speeds. Small to medium-scale wind turbines can be installed on-site to

generate electricity, either as a standalone system or in combination with solar power for a hybrid energy solution. Wind energy is particularly advantageous because it can produce power at night and during overcast conditions when solar panels may be less effective. Vertical-axis wind turbines (VAWTs) can be considered if space is a constraint, as they are compact and well-suited for industrial environments. Once installed, wind turbines have low operational costs and provide long-term energy savings.

6.9. Material Handling

The material handling is a very crucial part of the process as it ensures smooth, efficient and the safe movement of materials and equipment throughout the production process. For our facility, we will be utilizing machines and equipment such as forklifts and pallet jacks for material handling purposes. The following provides which material handling method is used between each stage of the manufacturing process with a justification on why this method is used and the material that will be moved.

Forklift Description

Dimensions: $3.2m \times 2.2m$ (*Length* $\times$ *Width*)

Space Occupancy: $3.2m \times 2.2m = 7.04 m^2$

A single forklift can lift up to 10000 kg (6 tons) and the coils weigh 2.5 tons.

This means that $6 \div 2.5 = 4$ *stations' output*. A single forklift can carry about 4 stations' output in both the drawing and galvanizing areas.



Figure 6.9.1: Forklift

1. Raw Material Storage → Drawing Machine

Material Handling Method we use to get undrawn steel coils for raw material storage to the drawing machines is with *pallet jacks*.

The pallet jacks will be sufficient to moved undrawn steel coils to the drawing machine. This will be done by one of the operators of each drawing machine, hence no extra workforce needed. An operator will go to collect an undrawn steel coil for raw material storage with a pallet jack, take it to the drawing machine of which they are the operator, and will return to collect another undrawn coil once they finished drawing the previous coil. This will require 20 pallet jacks as there are 20 drawing workstations, each to have their own pallet jack to avoid waiting among the operators.



Figure 6.9.2: Pallet Jack

2. Drawing Machine → Galvanizing Area

The material handling method to be used to move drawn recoiled steel wires to the galvanizing area is by means of *forklifts*.

Forklifts are very efficient since they are capable of moving and lifting heavy objects such as out drawn steel coils to the next station. The area of the galvanizing baths where the coils are placed before they are inserted into the machine (please see Figure 6.3.1), requires coils to be lifted into position, hence, we use forklifts for this operation and not do it manually or with pallet jacks. Operators from the galvanizing area collect these drawn coils from the drawing area with the forklift, hence no extra workforce is needed. As there are 20 drawing stations and from the forklift's capacity (4 stations' output), we get that 5 Forklifts will be needed. However, we don't want to use the forklifts to their full capacity as this places a lot of strain on them. We will hence use 8 forklifts for movement of material between drawing and galvanizing areas to ensure each forklift can move at least 2 stations' output to the galvanizing area.

3. Galvanizing Area → Netting Area

The material handling method to be used to move galvanized and recoiled steel wires to the galvanizing area is by means of *forklifts*.

Forklifts are very efficient since they are capable of moving and lifting these heavy coils with the added weight of the galvanized layer to the netting area. The part of the netting machine where the coils are placed before they are inserted into the machine (please see Figure 6.4.4), similarly to the galvanizing machine, requires coils to be lifted into position, hence, we use forklifts for this operation and not pallet jacks. Operators from the netting area, collect these recoiled galvanized steel wires from the galvanizing area and take it to the netting machine at which they operate, with the forklifts.

$$58 \text{ stations} \div 4 \text{ stations per forklift} = 14.5$$

This implies we need 15 forklifts for moving galvanized coils to the netting area.

4. Netting Area → Packaging Area

Pallets of netted rolls are moved to the packaging area with the material handling method being *pallet jacks*.

Pallet Jacks are used for to move large numbers of heavy netted rolls to the packaging area. Here operators will manually move rolls onto packaging machine. Operators from packaging area will collect netted rolls from the netting area and take them to their respective packaging stations they operate. This means, no extra workforce will be necessary. Since there are 97 packaging stations, we will be needing 97 pallet jacks to ensure each operator has their own pallet jack and no time will be wasted due to operators waiting for each other.

5. Packaging Area → Finished Goods Storage Area

The same operators will use their same *pallet jacks* to drop the packaged rolls of in the finished good storage area where the employees working in the finished good storage area will pack them neatly. There after the operators will head back to the netting area to collect a new batch of netted rolls to package.

7. Design Alternatives

7.1. Systematic Layout Planning

Activity Relationship

A total of 15 departments were decided to be used for the block layouts and master facility plan. The departments and their relationships is depicted in figure 7.1.1 below. The order of importance of closeness goes “A”, “E”, “I”, “O”, “U”, with “A” being absolutely important and “U” unimportant. There are no departments that should not be next to each other.

1. Raw Material Storage	A														
2. Drawing	A	U													
3. Galvanising	A	U	U												
4. Netting	A	U	U	U											
5. Packaging	A	U	U	U	U										
6. Small Order Storage	A	U	U	U	U	U									
7. Large Order Storage	O	U	U	U	U	U	U								
8. Office Department	O	U	U	U	U	U	U	O							
9. Cafeteria	I	U	U	U	U	U	U	O	E						
10. Reception	O	E	U	U	U	U	U	O	E	I					
11. Office Bathrooms	O	A	U	U	U	U	U	O	E	I	U				
12. Operator Locker Room & Bathrooms	E	O	U	U	U	U	U	O	E	I	U	U			
13. Forklift Parking	U	U	U	U	U	U	U	O	E	I	U	U	U		
14. Spares & Maintenance	U	U	U	U	U	U	U	O	E	I	U	U	U	U	
15. Segway Parking	U	U	U	U	U	U	U	O	E	I	U	U	U	U	U

Figure 7.1.1: Activity Relationship Chart

Dimensions

The dimensions of each department and the area this department takes up can be seen in table 7.1.1 below. The dimensions and areas include walkways and safety zones around the

machines and racks. There are three forklift parking areas that serve the WIP areas between drawing & galvanising, galvanising & netting and netting & packaging.

Table 7.1.1: Department Dimensions

Department	Dimensions (length x width)	Area (m ²)	Layout of machines/racks
Raw Material Storage	24m x 10m	240	5x5
Drawing	36m x 11m	396	8x4
	30m x 3.5m	105	4x1
Galvanising	101m x 81m	8181	7x1
Netting	112m x 49m	5488	7x7
	55m x 4.5m	248	10x1
Packaging	40m x 30m	1200	11x9
Small Order Storage	11m x 8m	88	6x5
Large Order Storage	32.5m x 8m	260	17x16
Office Departments	11m x 29m	319	-
Cafeteria	22m x 11m	242	-
Reception	11m x 5m	55	-
Office Bathrooms	2.5m x 2m	5	2x1
Operator locker rooms & bathrooms	22m x 10m	220	-
Forklift Parking	25m x 10m	250	-
Spares & Maintenance	30m x 11m	330	-
Segway Parking	11m x 4m	44	5x2 (10 segways)

From the activity relationship chart in figure 7.1.1 and the dimensions in table 7.1.1 above, the space relationship diagram can be drawn. This diagram can be seen in figure 7.1.2 below.

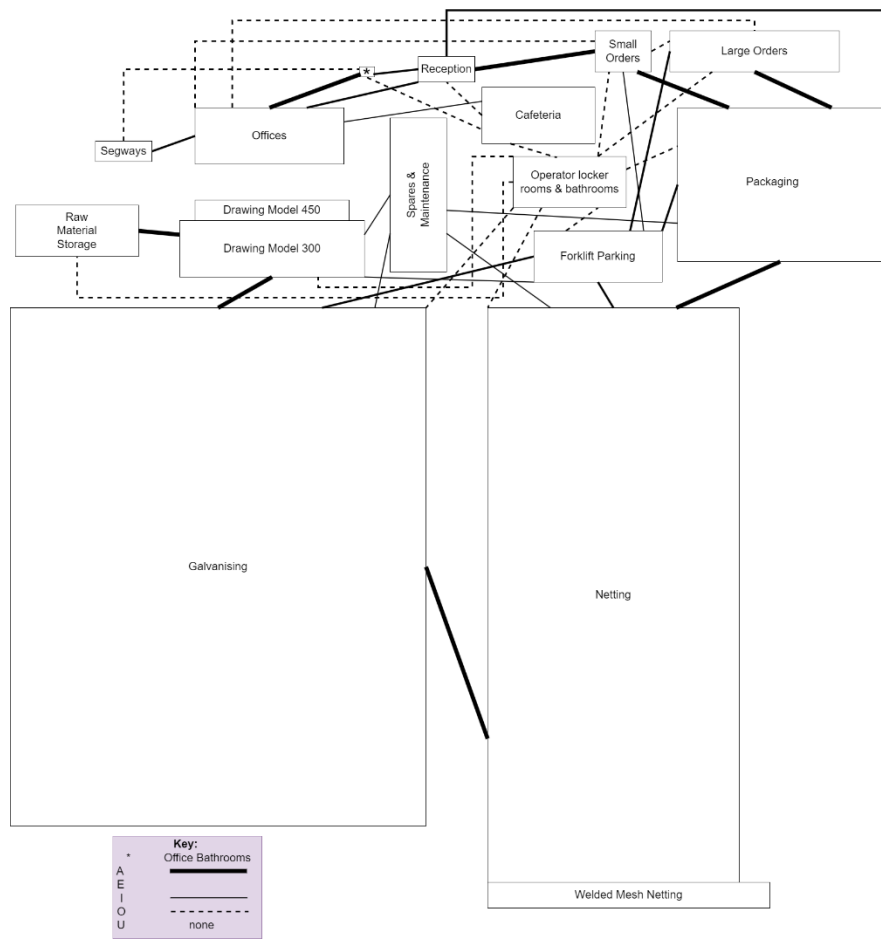


Figure 7.1.2: Space Relationship Diagram

7.2.Alternative Facility Layouts

Alternative 1

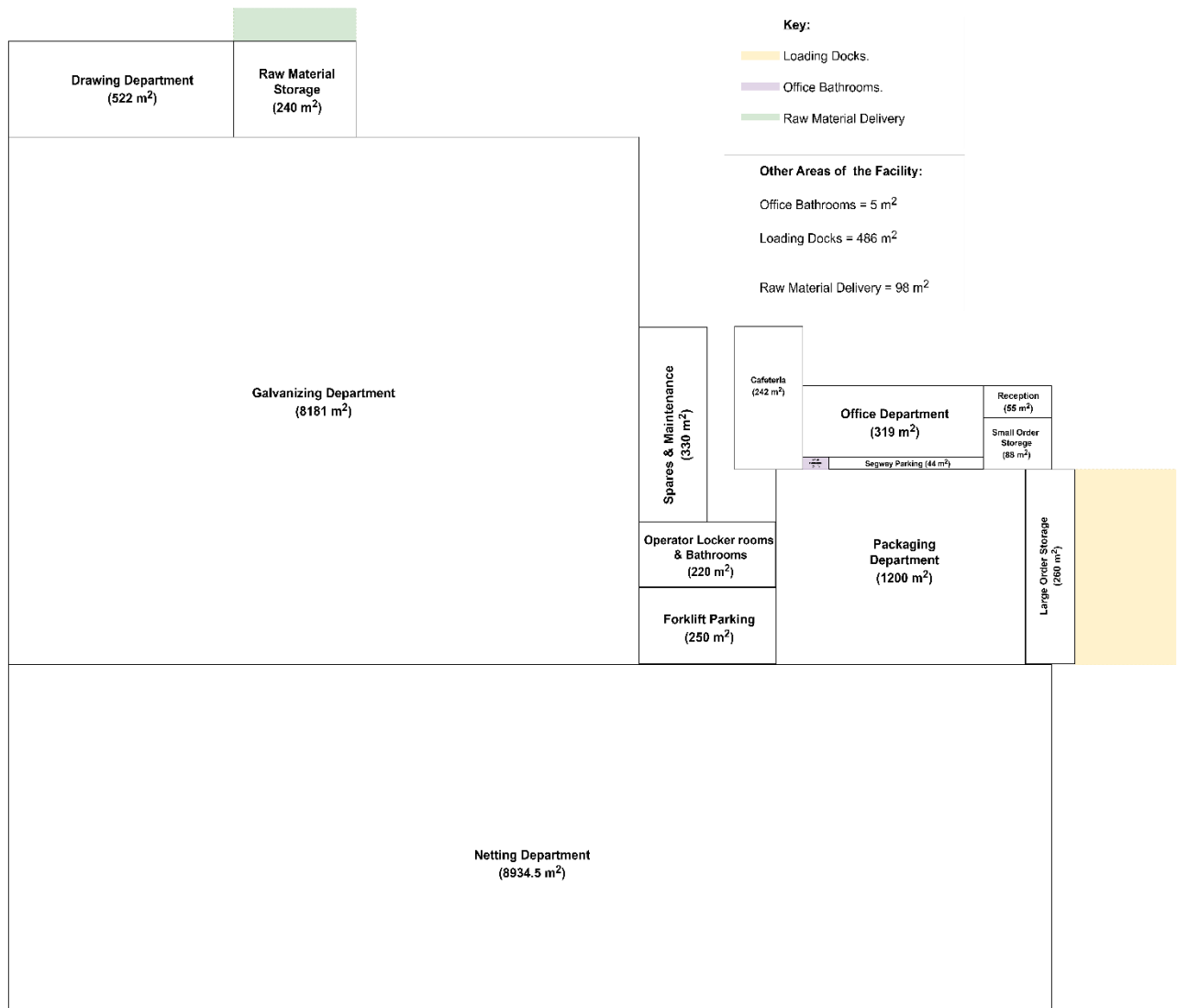


Figure 7.2.1: Alternative Block Layout 1

Alternative 2

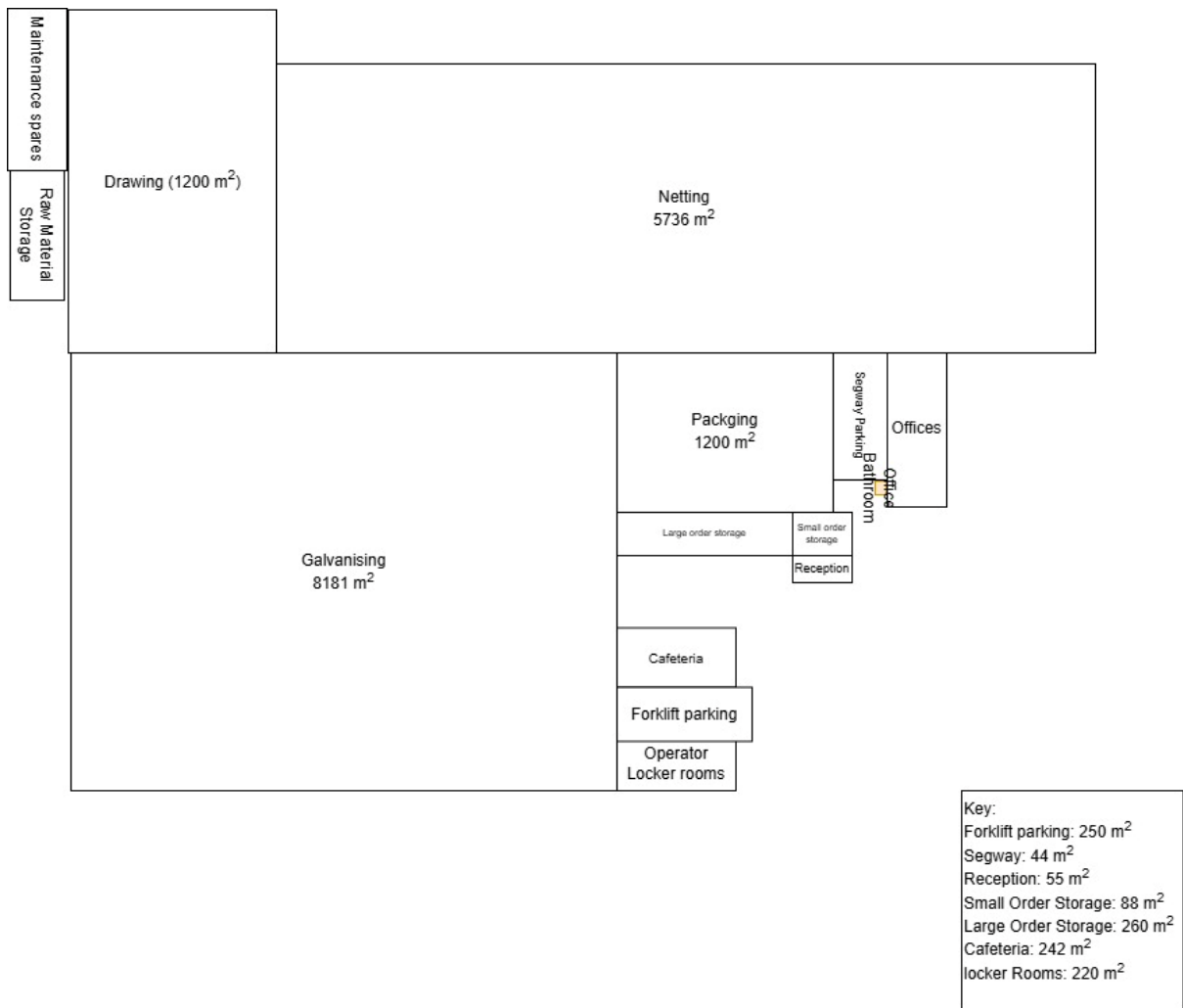


Figure 7.2.2: Alternative Block Layout 2

Alternative 3

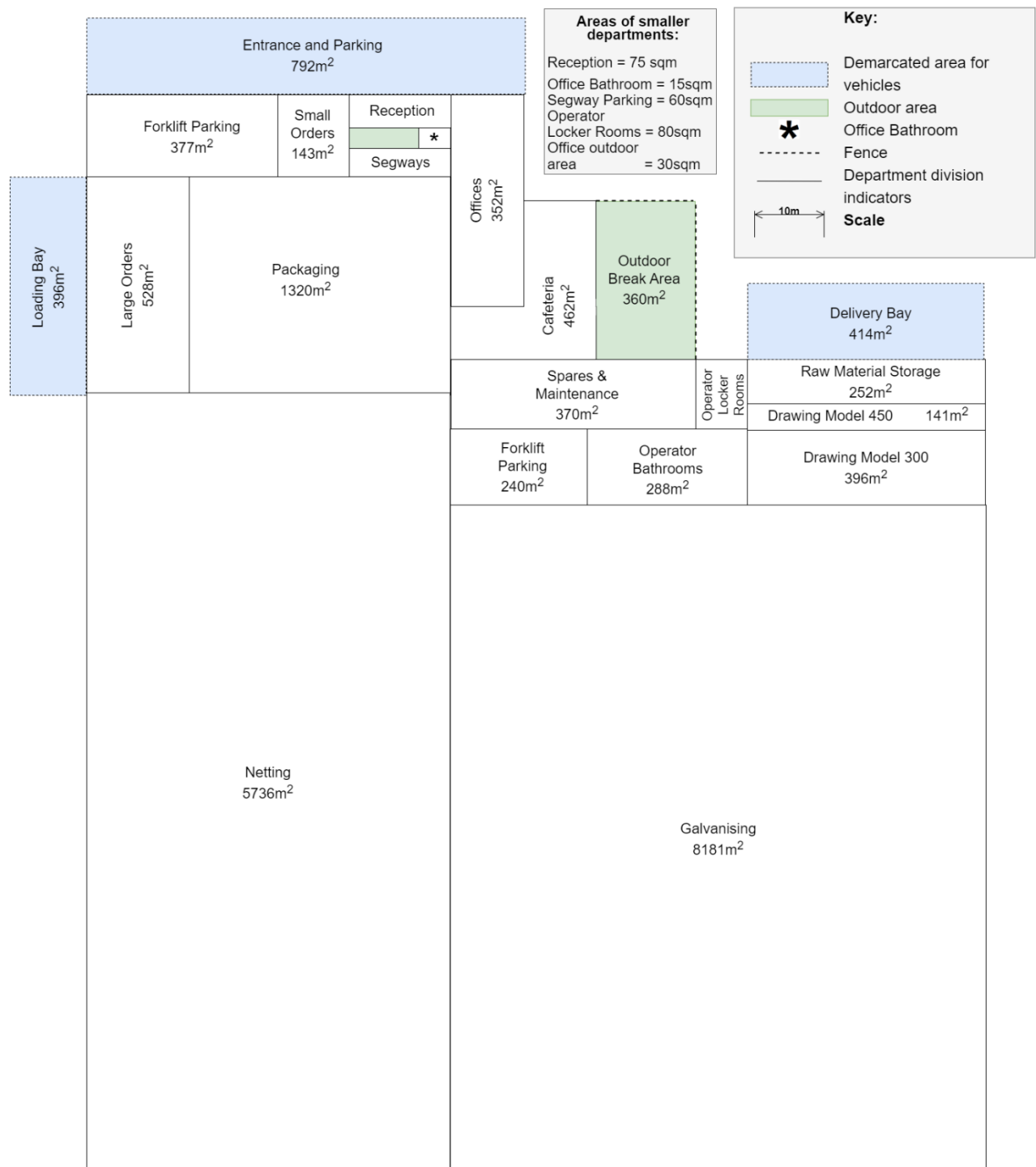


Figure 7.2.3: Alternative Block Layout 3

8. Alternative Evaluation

Table 8.1: Evaluation of alternative block layouts

Criteria	Alternative 1	Alternative 2	Alternative 3
Compliance with SLP	Yes	No	Yes
Compliance with allotted square footage per department	Yes	No	Yes
Advantages	Enough space for trucks to turn and drive Takes all needs into account	More square-like shape	Facility has a simple shape Facility takes all needs of the company into account
Disadvantages	Lots of sharp corners	No Drawing Department	Raw material delivery bay not well positioned for traffic flow
Total area occupied	18876m ²	18259m ²	19436m ²

9. Final Design

9.1. Plot Layout

The layout of the 50Ha plot is depicted below in figure 9.1.1. The facility is located in the top right corner of the plot.

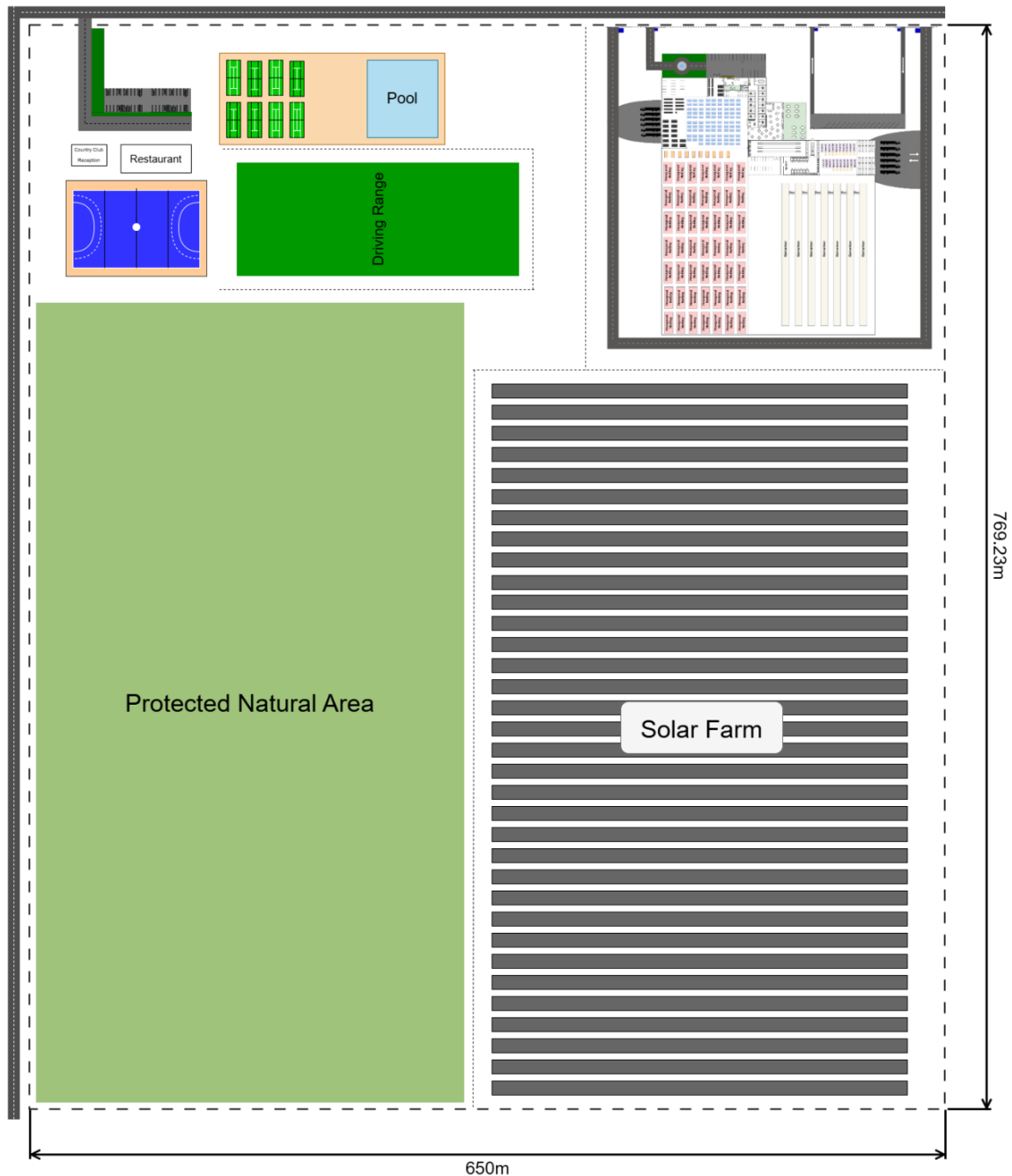


Figure 9.1.1: Plot of land

9.2. Master Facility Plan

For the Master Facility Plan (MFP), alternative layout 3 was mainly used. Some changes were made to the raw material delivery area. The flow of materials, processes, people and equipment is depicted in the adjusted block layout in figure 9.2.1 below.

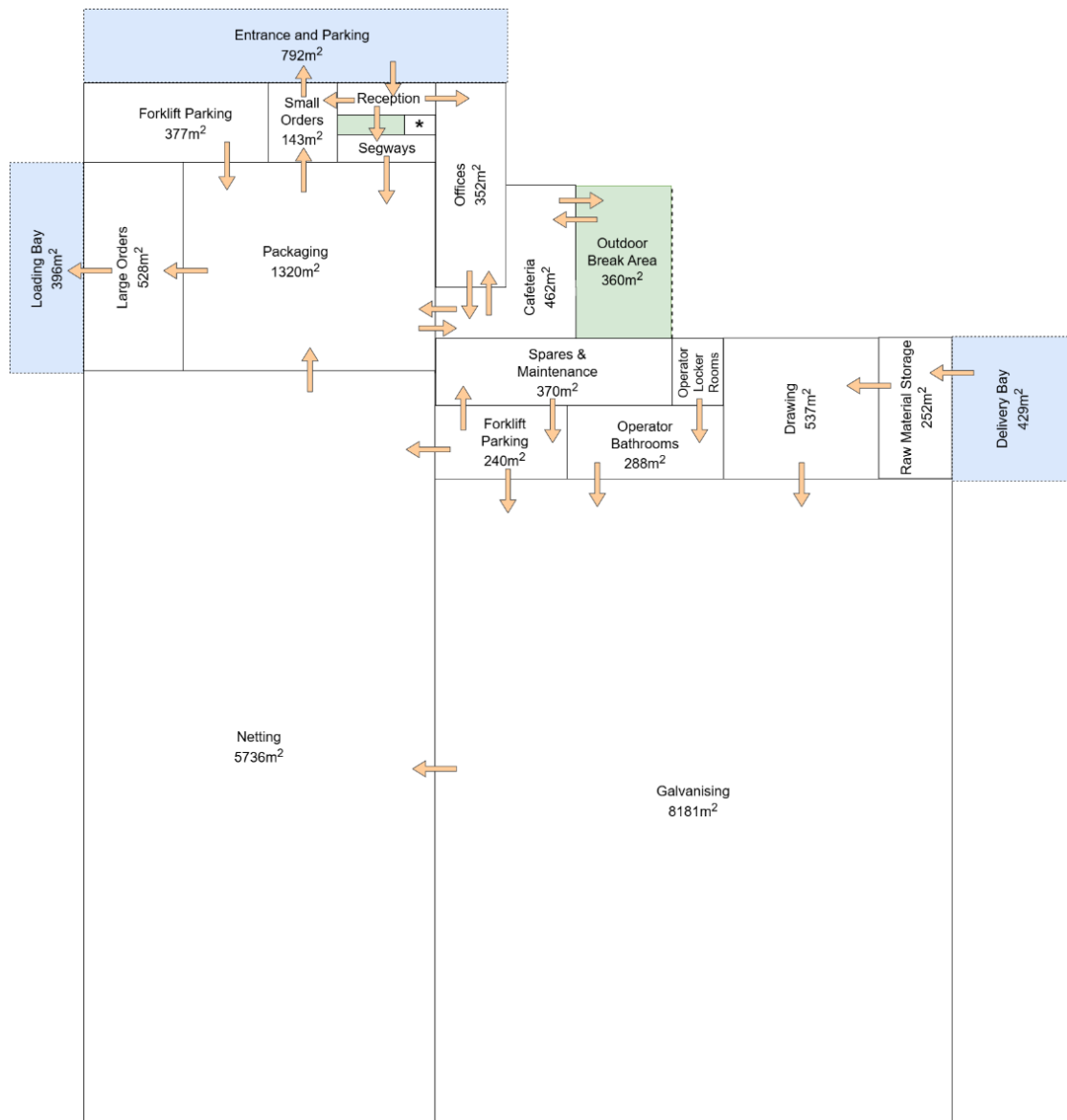
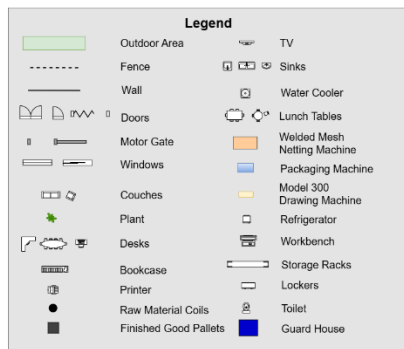
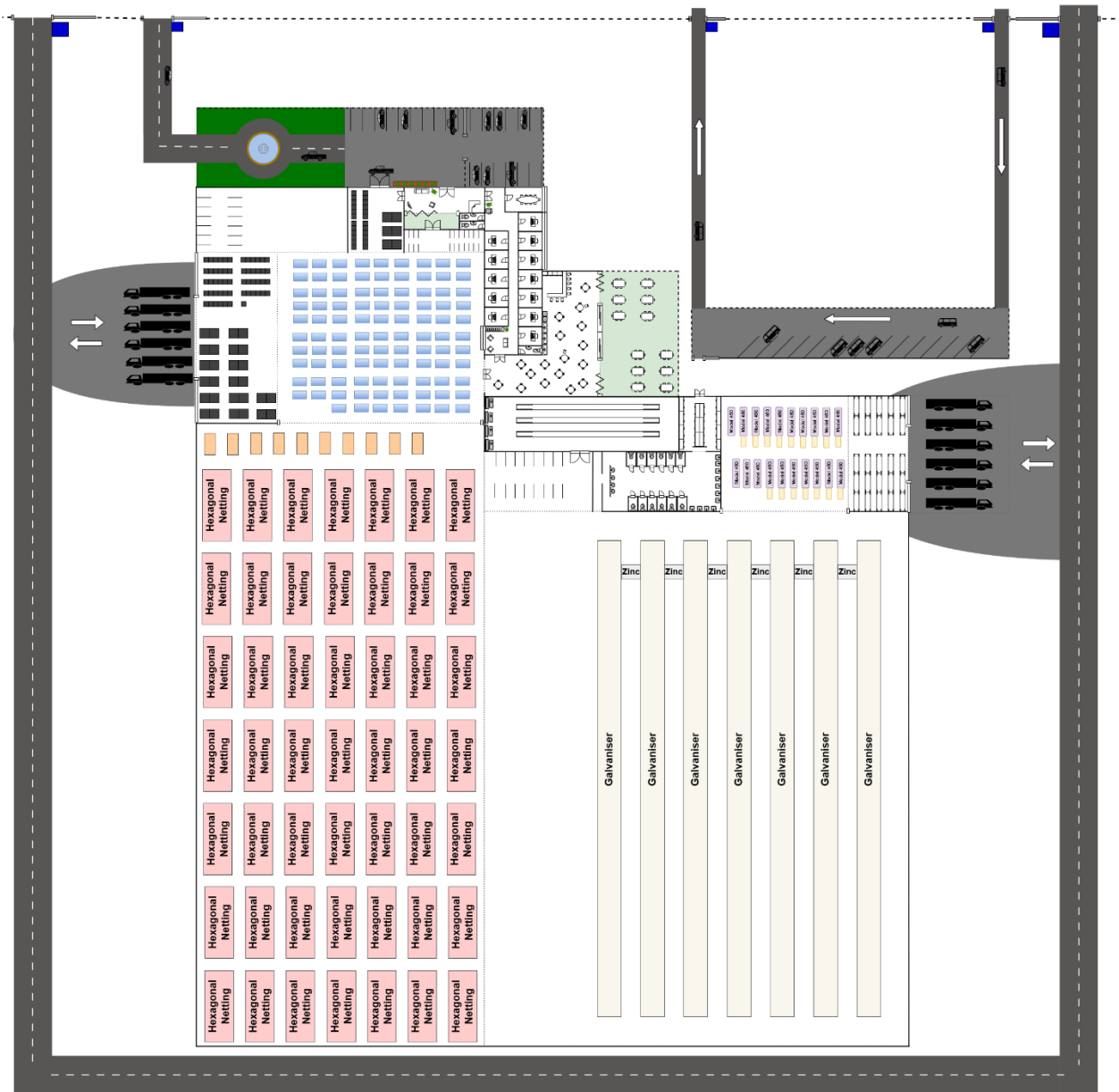


Figure 9.2.1: Final Block Layout with Flow Indicators

All departments were expanded by 10% to allow for future expansions Wiring.com is expecting. Roads, gates and parking for personnel and taxis were also added to the block layout and the final MFP can be seen below in figure 9.2.2.

Master Facility Plan for Wiring.com



Scale

Total Area 22 500m²
(excl. roads, delivery & loading bays)

Group 9:
Sheldon Farmer
Ofentse Mathebula
Karla van Dyk

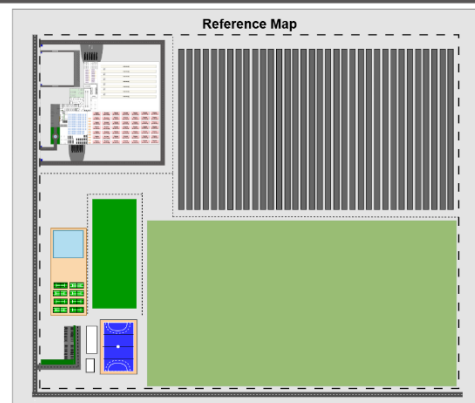


Figure 9.2.2: Master Facility Plan

10. Business Case

10.1. Capital Cost Estimates

A breakdown of the equipment costs can be seen in table 10.1.1 below.

Table 10.1.1: Equipment Costs

Equipment	Rate (R/equipment)	No. of equipment	Cost (R)
Storage Racks	50 000	28	1400000
Model 450 Netting	175 000	20	3500000
Model 300 Netting	100 000	15	1500000
Galvaniser	7 000 000	7	49000000
Hexagonal Netting	300 000	49	14700000
Welded Mesh Netting	150 000	10	1500000
Packaging Machine	75 000	97	7275000
Workbenches	20 000	4	80000
Forklifts	400 000	24	9600000
Pallet Jacks	35 000	120	4200000
Total Cost			92 755 000

The total capital cost estimates (including the equipment costs calculated above) is calculated and can be seen in table 10.1.2 below. Operational and procurement costs are not taken into account as it does not form part of the facility costs.

Table 10.1.2: Capital Cost Estimates

Component	Rate (R/ m ²)	Total m ²	Cost (R)
Building	5 000	21 625	108 125 000
Offices	10 000	350	3500000
Cafeteria	10 000	460	4600000
Parking & Roads	800	6 145	4916000
Restrooms	7 500	385	2887500
Electrical	1 500	22 000	33000000
Lighting	750	21 625	16218750
Ventilation	1 000	21 625	21625000
Fire Safety	300	22 000	6600000
Equipment	-	-	92755000
Total Cost			344112500

10.2. Risks and Mitigations

The following table outlines the potential risks identified for the project along with their impacts and proposed mitigation strategies.

Table 10.2.1: Risks & Mitigation Strategies

Risk	Impact	Mitigation Strategy
Construction delays – Late site preparation or permitting	Project timeline slip; increased labour/equipment costs	Early engagement with local authorities for fast-track permitting; Include

		10% time contingency in schedule
Cost overruns – Material price volatility (steel, zinc)	Budget exceedance; ROI reduction	Lock-in key material prices via forward contracts; Maintain 10% budget contingency reserve
Equipment selection risk – Delivery lead-times longer than planned	Delayed commissioning; lost production days	Pre-qualify multiple suppliers; Negotiate penalty clauses for late delivery
Supply-chain disruption – Raw-material (steel-coil) shortages or transport issues	Production stoppage; inability to meet customer demand	Maintain 3 days' raw-material stock on-site; Develop secondary supplier relationships
Quality failures – Scrap rate higher than the assumed 4% in Operation A	Increased waste; extra handling/storage space required	Implement in-process quality checks; Train operators and establish quick-response scrap removal
Health, Safety & Environment (HSE) – Incident in galvanizing bath or waste handling	Regulatory fines; work stoppage	Provide PPE, safety training and emergency drills

10.3.Schedule

The image below is a high-level project schedule summarizing the start, finish, and duration of each major phase:

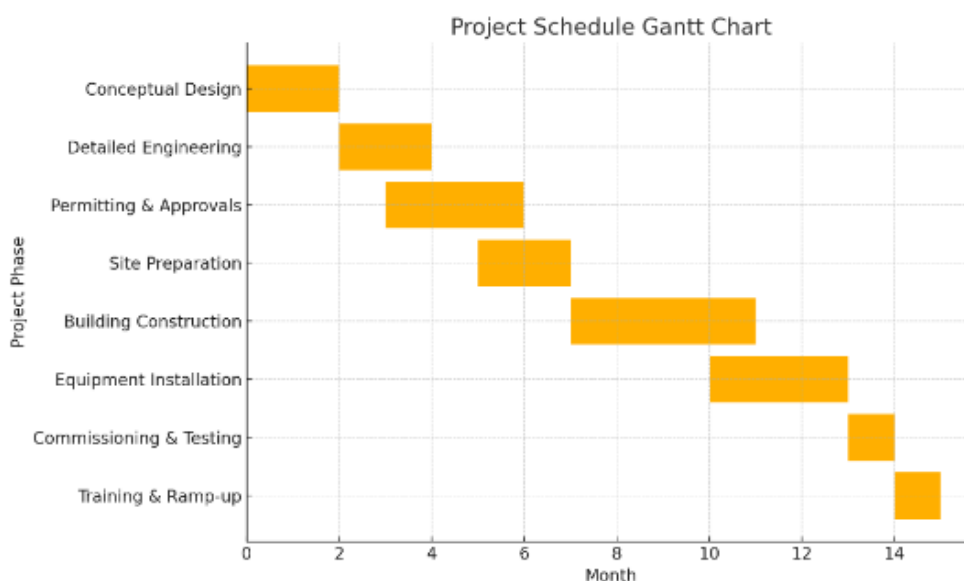


Figure 10.2.2: Schedule

Appendix A (Calculations)

Drawing

Product	Height of roll (m)	Aperture (mm)	Wire Diameter (m)	Roll length (m)	Weight per roll (kg)	Galvanized finish	Demand per annum (tons)
Bird Netting	0.9	13	0.0006	50	22.5	Light	4000
Chicken Netting	1.8	25	0.0009	50	41.9	Light	5500
Jackal Netting	0.6	75	0.0018	50	19.5	Light	2500
Pig Netting	0.9	90	0.0018	30	14.7	Light	8000
Welded Mesh	2	50	0.0025	30	70	Heavy	6000
Totals							26000

Weight of steel wire coil	2500
Length of raw material coil (m)	7000
Original diameter (m)	0.005

	Length of coil on exit (m)	Weight per meter (kg)	Metres per roll	Rolls per annum	m needed per annum	km needed per annum
Bird Netting	486111.11	0.0051	4375.00	17777.78	7777777.8	77777.7778
Chicken Netting	216049.38	0.0116	3620.99	131264.92	475308642	475308.642
Jackal Netting	54012.35	0.0463	421.30	128205.13	54012345.68	54012.34568
Pig Netting	54012.35	0.0463	317.59	544217.69	172839506.2	172839.5062
Welded Mesh	28000.00	0.0893	784.00	85714.29	67200000	67200
Total	838185.19	0.1986	9518.88	1067179.80	1547138271.60	1547138.27

Days worked per day						
	After draw	Before draw	With 4% scrap allowance (after draw)	With 4% scrap allowance (before draw)		
	m per day	km per day	m per day	km per day	m per day	km per day
Bird Netting	3535353.535	3535.353535	50.90909091	3682659.933	53030.30303	53.03030303
Chicken Netting	2160493.827	2160.493827	70000	2250514.403	72916.66667	72.91666667
Jackal Netting	245510.6622	245.5106622	31.81818182	255740.2731	33143.99939	33.14399939
Pig Netting	785634.119	785.634119	101818.1818	818368.8739	106060.6061	106.0606061
Welded Mesh	305454.5455	305.4545455	763636.3636	318181.8182	79545.45455	79.54545455
Total	7032446.689	7032.446689	330909.0909	7325465.301	344696.9697	344.6969697

Daily Production Requirement: Drawing Speed (m/s)			
	Drawing Speed machine 1	Length one machine draws per day	No. of machines needed
Bird Netting	3682659.933	15	20
Chicken Netting	2250514.403	15	12
Jackal Netting	255740.2731	20	1
Pig Netting	818368.8739	20	2
Welded Mesh	318181.8182	20	1
Total	7325465.301	90	36
		*use for 7hrs per day (1hr cleaning and setup time)	

No. of raw material coils needed	49,242,424
No. of raw material coils in storage (3 days)	147,727,272

	Height of roll (m)	Roll length (m)	Material thickness (m)
Bird Netting	0.9	50	0.0008
Chicken Netting	1.8	50	0.0011
Jackal Netting	0.6	50	0.002
Pig Netting	0.9	30	0.002
Welded Mesh	2	30	0.0027

Galvanisation

Total metal to be galvanised per day (km)		7032,446689		pi		3,141592654	
Zinc density (g/cm³)		7.14					
Zinc mass per unit area (g/m²)		357					

Wire diameter (m)	Surface area per meter (m²)	Zinc mass per meter (g)	Zinc mass per km (kg)
0,0006	0,00204	520,5106947	520,5106947
0,0009	0,00298	760,7464	760,7464
0,0018	0,00581	1481,453516	1481,453516
0,0018	0,00581	1481,453516	1481,453516
0,0025	0,00801	2042,003495	2042,003495
Total			6286,16762

Galvaniser dimensions:			
Length (m)	100		
Width (m)	5		
Depth (m)	3		
Galvanising speed (kg/h)	600		

	yearly requirement (kg)	daily requirement (kg)	hourly requirement (kg)	machines needed
Bird Netting	4000000	18181,81818	2597,402597	1,082251082
Chicken Netting	5500000	25000	3571,428571	1,488095238
Jackal Netting	2500000	11363,63636	1623,376623	0,676406926
Pig Netting	8000000	36363,63636	5194,805195	2,164502165
Welded Mesh	6000000	27272,72727	3896,103896	1,623376623
	26000000	118181,8182	16883,11688	7

Zinc ingot weight (kg)			
Daily demand (m)	25		
Bird Netting	3682659,933	0,0003	6941,650434
Chicken Netting	2250514,403	0,00045	6363,179565
Jackal Netting	255740,2731	0,0009	1446,177174
Pig Netting	818368,8739	0,0009	4627,766956
Welded Mesh	318181,8182	0,00125	2498,994156
	7325465,301		21877,76828

No. of ingots needed daily	302,51713		
No. of ingots needed for work week	1512,58563		
Length of pallet (m)	1		
Width of pallet (m)	0,5		
Height of pallet (m)	0,4		
No. of layers in pallet	10		
No. of slabs on a pallet layer	8		
Area for 5 days storage (sqm)	9,453660185		

*work with 7hrs per day of galvanising time (1hr setup and cleaning time)			
gavlanised coating (kg/m²)	daily zinc demand (kg)		
0,3	2082,49513		
0,3	1908,953869		
0,3	433,8531521		
0,3	1388,330087		
0,7	1749,295909		
	7562,928148		

Finished Goods Storage

Product	Height of roll (m)	Aperture (mm)	Wire Diameter (m)	Roll length (m)	Roll Width (m)	Weight per roll (kg)	Galvanized finish	Customer requirements per annum (kg)
Bird Netting	0.9	13	0.0006		50	0.25	22.5 Light	4000000
Chicken Netting	1.8	25	0.0009		50	0.28	41.9 Light	5500000
Jackal Netting	0.6	75	0.0018		50	0.37	19.5 Light	2500000
Pig Netting	0.9	90	0.0018		30	0.29	14.7 Light	8000000
Welded Mesh	2	50	0.0025		30	0.34	70 Heavy	6000000
Totals								

Rolls produced per day		Area needed for one roll		Total area of daily rolls (sqm)		Weekly storage space needed (sqm)		Pallets	
Bird Netting	808,080,801	81	0.225	4,55625	31,89375	5,0625			
Chicken Netting	596,6587112	60	0.504	7,36	52,92	3,75			
Jackal Netting	582,7505828	59	0.222	3,2745	22,9215	3,6875			
Pig Netting	2473,716759	248	0.261	16,182	113,274	15,5			
Welded Mesh	389,6103896	39	0.68	6,63	46,41	2,4375			
Totals	4850,817251	487	1,892	38,20275	267,41825	30,4375			

Rolls produced per day		Area needed for one roll		Total area of daily rolls (sqm)		Weekly storage space needed (sqm)		Pallets	
Bird Netting	808,080,801	728	0.225	40,95	81,9	11,375			
Chicken Netting	596,6587112	537	0.504	67,662	135,324	8,390625			
Jackal Netting	582,7505828	525	0.222	29,1375	58,275	8,203125			
Pig Netting	2473,716759	2227	0.261	145,31175	290,6235	34,796875			
Welded Mesh	389,6103896	351	0.68	59,67	119,34	5,484375			
Totals	4850,817251	4366	1,892	342,73125	685,4625	272,875			

Appendix B (Minutes of Meetings)

Minutes of Meeting (3 March 2025)

Time: 10:00-11:00
Location: Engineering Campus Mini-SS
Meeting called to order by: Karla van Dyk
Minutes of meeting taken by: Karla van Dyk

Attendance:

Sheldon Farmer

Karla van Dyk

Absent:

Ofentse Mathebula

Reason:

Did not have transport to the meeting

Agenda:

1. Group member introduction
2. Talk-through of assignment
3. Initial job allocation

Decisions:

1. Initial work duties (due on 11 March 2025) were divided as follows:

Person Responsible	Duties
Sheldon Farmer	• Research wire drawing machines • Research wire braiding machines • Calculations about demand • Workstation layouts of drawing and netting process
Ofentse Mathebula	• Research packaging and packaging process • Workstation layout for packaging process • Research safety and factory requirements • Research delivery truck capacities
Karla van Dyk	• Research sustainable energy options • Research galvanising process • Calculations about galvanising process • Calculations about raw material

2. All members agreed to attend the *class session on 4 March 2025 to work on assignment and ask any questions that might arise.

**Mr Mathebula did not end up attending the class session.*

3. Ms Van Dyk will create a document for everyone to edit on and will also format this document.
(link:https://docs.google.com/document/d/1atGOHg7HVeIz1RXGH4uANZLSe_n94bHxI8xfl-fTWPY/edit?usp=sharing)
4. If it were to rain during future meetings, either Mr Farmer or Ms Van Dyk will provide transport for Mr Mathebula.

Next Meeting:

The next meeting is scheduled for Wednesday 12 March 2025 at 11:00 at the Main Campus SS. Notice for this meeting was sent out on Friday 7 March 2025 via the Whatsapp group.

Minutes of Meeting (12 March 2025)

Time: 11:00-12:00
Location: Main campus SS
Meeting called to order by: Sheldon Farmer
Minutes of meeting taken by: Karla van Dyk

Attendance:

Sheldon Farmer
Ofentse Mathebula
Karla van Dyk

Agenda:

1. Check-in on progress of group members
2. Next phase of work allocation
3. Small orders

Decisions:

1. Work duties (due on 19 March 2025) were divided as follows:

Person Responsible	Duties
Sheldon Farmer	• Initial workstation layout for drawing process • Initial workstation layouts of netting process
Ofentse Mathebula	• Packaging calculations • Initial workstation layout for packaging process
Karla van Dyk	• Research sustainable energy options • Initial workstation layout for galvanising • High-level process flow drawing

2. A small order is defined as an order that can be picked up by customers themselves (10 rolls or less) and it constitutes 10% of total production.
3. Start writing report after next week's meeting

Next Meeting:

Next meeting is scheduled for Wednesday 19 March 2025 at 11:00 at the Main Campus SS.

Minutes of Meeting (19 March 2025)

Time: 11:00-12:00
Location: Main campus SS
Meeting called to order by: Karla van Dyk
Minutes of meeting taken by: Karla van Dyk

Attendance:

Sheldon Farmer
Ofentse Mathebula
Karla van Dyk

Agenda:

1. Check-in on progress of group members
2. Assess what needs to be done before submission
3. Internal submission deadline
4. Material handling

Decisions:

1. All members progressed well with their responsibilities from the previous week
2. Work duties (due on 23 March 2025) were divided as follows:

Person Responsible	Duties
Sheldon Farmer	• Write background for report • Complete workstation layout • Write detailed design specifications for drawing and netting
Ofentse Mathebula	• Write design objectives for report • Complete workstation layout • Complete calculations for packaging • Write detailed design specifications for packaging
Karla van Dyk	• Write assumptions and constraints for report • Complete workstation layouts • Write detailed design specifications for galvanising, raw material storage and finished good storage • Format report and label figures

3. Report must be completed by Sunday 23 March 2025
4. Need to research material handling specifically between the drawing machines and galvanisers. Particularly look at some sort of forklift.

Next Meeting:

Next meeting is scheduled for Wednesday 26 March 2025 at 11:00 at the Main Campus SS.
This meeting will be used to discuss what still needs to be done for the final submission.

Minutes of Meeting (26 March 2025)

Time: 11:00-11:30
Location: Online (MS Teams)
Meeting called to order by: Karla van Dyk
Minutes of meeting taken by: Karla van Dyk

Attendance:

Sheldon Farmer
Ofentse Mathebula
Karla van Dyk

Agenda:

1. Establish what needs to be done for the final submission
2. Divide work accordingly and set internal deadlines
3. Establish which departments will be used

Decisions:

1. Sections to be added to report are Material Handling, Design Alternatives, Alternative Evaluation, Final Design and Business Case
2. Work duties (due on 2 April 2025) were divided as follows:

Person Responsible	Duties
Sheldon Farmer	• Compile list of different departments and rank them against each other
Ofentse Mathebula	• Research personnel movement solutions for within the plant
Karla van Dyk	• Research and establish personnel requirements for plant operation

3. All members will complete their own sub-section of material handling section
4. SLP and AHP will be used to design and evaluate alternative layouts

Next Meeting:

Next meeting is scheduled for Wednesday 2 April 2025 at 11:00 at the Main Campus SS.

Minutes of Meeting (2 April 2025)

Time: 11:30-12:30
Location: Main Campus SS
Meeting called to order by: Karla van Dyk
Minutes of meeting taken by: Karla van Dyk

Attendance:

Sheldon Farmer
Karla van Dyk

Absent:

Ofentse Mathebula

Reason:

Had to sort out issues with bursary

Agenda:

1. Feedback of assignment 4 (first submission of report)
2. Activity Relationship Chart
3. Future internal deadlines

Decisions:

1. All feedback of the first assignment was worked through and each member knows what has to be fixed, added and removed.
2. The activity relationship chart was set up and closeness relations were assigned to all departments. This diagram will be used to draw up SLP and do block layouts from.
3. The following internal deadlines for the remainder of the work was decided upon and agreed to by all members via Whatsapp:

21 April 2025	Alternative Block Layouts
22 April 2025	Evaluation of Alternatives
28 April 2025	Master Facility Plan
1 May 2025	Business Case
2 May 2025	Everything that has to be changed/adapted

Next Meeting:

Next meeting is scheduled for Tuesday 22 April at 10:00 online on MS Teams.

Minutes of Meeting (22 April 2025)

Time: 15:45-16:30
Location: Online (MS Teams)
Meeting called to order by: Karla van Dyk
Minutes of meeting taken by: Karla van Dyk

Attendance:

Sheldon Farmer
Ofentse Mathebula
Karla van Dyk

Agenda:

1. MFP Evaluation
2. Future internal deadlines

Decisions:

1. The alternative block layouts were evaluated and compared against each other and a design for the MFP was decided on. All group members were assigned different parts of the MFP to draw
2. The deadline for MFP completion was extended by one day and the Business Case deadline by two.

Next Meeting:

Next meeting is scheduled for Tuesday 29 April at 11:00 online on MS Teams.

Minutes of Meeting (29 April 2025)

Time: 11:00-11:30
Location: Online (MS Teams)
Meeting called to order by: Karla van Dyk
Minutes of meeting taken by: Karla van Dyk

Attendance:

Sheldon Farmer
Ofentse Mathebula
Karla van Dyk

Agenda:

1. Evaluation of MFP
2. Division of remaining work

Decisions:

1. The MFP was evaluated and small changes were noted to be made
2. The Business Case and the Material Handling sections were divided among the members of the group for completion by 2 May
3. All corrections and changes to be done by 4 May the in the afternoon to be ready for submission in the evening.

Next Meeting:

No next meeting will be held as the assignment is to be submitted on 4 May 2025.